Custom System Calls (xv6)

- Implemented 3 system calls:
- getppid, sleep2, getprocsinfo

getppid

- Returns parent process ID
- Uses myproc() and parent pointer

sleep2

- Pauses process for given time
- Uses ticks and sleep mechanism

getprocsinfo

- Displays all active processes
- Prints PID, state, memory size

SCREENSHOT:

```
Running getppid...
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ test_getppid
My PID: 3
Parent PID: 2
$
Running sleep2...
test_sleep2
Start

Running procs...
test_procs
End
$ PID   STATE   SIZE
1       2       16384
2       2       20480
5       4       16384
$
```

System Call Flow

- User → user.h → usys.pl → syscall.c → sysproc.c → Kernel